Application: Brake Rotor Temperature Measurements

The Challenge

Brakes turn the kinetic energy of a vehicle's motion into heat, which brake rotors must then dissipate to the atmosphere. With continued use, the brake system gets hotter and hotter, and as the heat builds up, the vehicle may experience a loss of stopping power and potentially warped brake rotors. Therefore, measuring brake rotor temperature is important in the development of optimized braking systems. However, obtaining this data in the real-world is difficult, not only due to the extreme temperatures, but also due to packaging constraints. Thermocouples are needed for accurate measurement across a wide temperature range, but these sensors need both a power source and a data acquisition system, requiring wires, which cannot be used on a rotating system.

System Requirements

- ✓ Wireless: The measurement system must wirelessly transmit the data from the thermocouples to a data acquisition system.
- ✓ Compact: The system must be able to be packaged into a vehicle wheel hub.
- ✓ Low power consumption: The power source must be able to sustain long term measurements of two thermocouples, and power the electronics to transmit the data wirelessly.
- ✓ Heat resistant: The sensor system must be able to withstand the high temperatures of the wheel hub environment.

The TECAT Solution

TECAT custom designed a wireless temperature measurement system based on the WISER product that could be packaged inside the wheel hub. The remote sensing unit consists of a standard enclosure, shown here, with two K-type thermocouples, a lithium poly battery and remote unit electronics. This standard enclosure mounts to a custom designed mounting plate for installation in a specific wheel hub. The mounting plate is the only unique piece required for each application, minimizing cost for use across multiple wheel hubs.

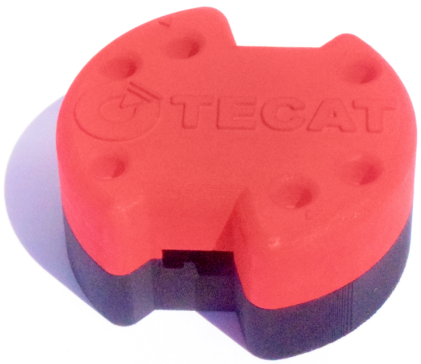


Figure 1 Standard Remote Enclosure

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